Development and Characterisation of Eggshell–Agar Biocomposites for Sustainable Interior Applications



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Eggshell valorisation is established

WHERE EGGSHELL WASTE CURRENTLY GOES

Application	Matrix	Reference
Cement replacement	Portland clinker	Teara et al. 2019
Laterite brick stabilisation	Cement	Adogla et al. 2016
Asphalt binder modification	Bitumen	Razzaq & Yousif 2017
Soil stabilisation	Lime substitute	Soundara & Vilasini 2015
Eco-ceramic wall tile	Ceramic body	Vilarinho et al. 2022
Particulate filler	PLA, epoxy, PVC, PU	Owuamanam & Cree 2020

In every case the matrix is petrochemical, clinker-based or ceramic.

William & Owen 1995 · Torres et al. 2010 · Shafiei et al. 2019

WHY THE FIELD RETURNS TO IT

- ~94% calcium carbonate
- Porous, protein-bound crystalline structure
- Specific gravity 0.85–2.66
- Continuously generated, landfilled at scale

The material argument is settled. The open question is what binds it.

Agar is characterised — but only as a film

WHAT IS KNOWN

Hernández et al. (2022) mapped 32 agar–glycerin–water formulations by design of experiments and released the dataset openly. Across the film literature, glycerol raises elongation and lowers tensile strength.

WHAT IS NOT

That work concerns unfilled thin films in uniaxial tension, for packaging. Agar has not been characterised as the binder of a mineral-filled cast solid in bending — the loading case for an interior surface.

THE SHIFT THIS STUDY MAKES

—	Film literature	This study
Form	Thin cast film	Cast solid, 20–50 mm
Filler	Unfilled	Mineral, 66% of solids
Loading	Tension	Compression, bending
Use	Packaging	Interior surfaces

The compositional logic carries over. The form, the filler and the loading case do not.

The gap is documented, not asserted

AN OPEN COMMONS EXISTS

Materiom publishes biomaterial recipes under CC-BY-SA, excludes petrochemical ingredients, archives each as prior art, and indexes property data with citable DOIs. The infrastructure is built.

WHAT THAT MEANS HERE

The nearest analogue is the mussel shell–sucrose ceramic: a shell mineral in a biological binder. Nothing couples a shell mineral to an agar binder, and nothing reports compressive, flexural, ignitability and hygric data for one formulation family.

4

recipes with published property data

agar bioplastic
gelatin bioplastic
mycelium–walnut shell
mussel shell–sucrose ceramic

A protocol and a property set entering a commons four recipes deep.

materiom.org/data, accessed October 2026

Aim and research questions

AIM

To document a reproducible production protocol for a fully bio-based eggshell composite, and to characterise its mechanical, thermal and hygric response against conventional and bio-based interior materials.

SCOPE

- Five formulations, two mechanical programmes
- Three specimens per formulation
- ASTM D790-17, ISO 11925-2, ASTM D570
- Single laboratory, ambient curing, no ageing

RESEARCH QUESTIONS

Q1 Which formulation variable governs compressive strength?

Q2 What is the effect of plasticiser content on flexural behaviour?

Q3 How does it compare with conventional interior materials?

Q4 How does it compare with other bio-based composites?

Q5 Is compressive strength the appropriate criterion at all?

Constituents and their function

Constituent	Origin	Function	Relevant property
Eggshell powder	Food waste	Mineral filler; carries load	94% $CaCO_3$; porous structure
Agar	Red seaweed	Continuous binding phase	Thermoreversible gel
Glycerol	Vegetable oil	Plasticiser; sets ductility	Humectant; low vapour pressure
Water	—	Carrier; sets solids packing	Fully evaporated on drying
Shellac	Kerria lacca	Hydrophobic surface finish	Biodegradable film-former

Water is not a component of the finished material. It is the process variable that determines how densely the solids pack.

Gelatine, starch and vinegar were trialled during development and are not part of the reported formulations.

Protocol I — mineral fraction

01 **Collection and cleaning**

Single local bakery, over one year. Membrane and organic residue removed.

02 **Thermal treatment**

120 °C for 20 min. Decontamination and drying, not sterilisation: sterility was not verified.

03 **Cooling and drying**

To ambient, then dried until no further mass loss.

04 **Milling**

Laboratory mechanical grinder, to a visually homogeneous powder.

05 **Sieving**

150 µm mesh; passing fraction only, consistent with polymer-composite practice.



120 °C
20 min

150 µm
sieve aperture

Protocol II — binder phase

01 **Pre-mixing**

Water and glycerol to a homogeneous liquid before any solids enter.

02 **Agar dispersion**

Added incrementally under continuous stirring. Bulk addition forms agglomerates that do not redisperse.

03 **Dissolution**

Heated to 95 °C and held until the agar is fully in solution.

04 **Controlled cook**

Reduced to 80 °C and held, stirring maintained, while viscosity develops.

Stirring was manual. Shear rate and mixing energy are not controlled quantities.

THERMAL REGIME

95 °C

dissolution set point

80 °C

controlled cook

55 min

total heating time

Two-stage regime follows the 70–90 °C band reported for agar preparation.

Protocol III — compositing, casting, drying

01 **Mineral incorporation**

Added incrementally into the binder at 80 °C. Earlier addition lets the mineral sediment out.

02 **Casting**

Silicone moulds, levelled. Silicone releases without a parting agent. 50 mm cubes for compression, flat moulds otherwise.

03 **In-mould setting**

Four days.

04 **Ambient consolidation**

Demoulded, then seven days at ~23 °C, shielded from sun and air movement. No oven at any stage.

DISCLOSURE — DIMENSIONAL LOSS

The composite shrinks appreciably as water evaporates. Cross-sections were measured at test time, but post-drying dimensions were not logged. Dimensions quoted here are mould dimensions.

23 °C · 11 days · no oven

Anyone applying this protocol should record geometry after drying, not at casting.

Formulation matrix

ID	Test	Water g	Shell g	Agar g	Glycerol g	Solids g	Shell : binder	Solids : water	Glycerol % solids
C1	Compression	325	80	35	5	120	**2.00**	0.37	4.2
C2	Compression	310	90	40	5	135	**2.00**	0.44	3.7
C3	Compression	280	110	45	10	165	**2.00**	0.59	6.1
F1	Flexure	450	190	70	60	320	1.46	0.71	18.8
F2	Flexure	480	190	70	30	290	1.90	0.60	10.3

COMPRESSION SERIES

Shell : binder is held at exactly 2.00. The only variable that changes is solids concentration, 0.37 to 0.59.

FLEXURAL SERIES — CONFOUND

Glycerol was halved but batch mass was fixed at 770 g, so water rose to compensate and the other ratios moved with it. The plasticiser effect is not isolated.

Three specimens per formulation. Values are means; individual specimen values were not retained, so standard deviations are not available.

Test programme

Property	Standard	Specimen	n	Measured
Compressive behaviour	Cube compression	50 mm cube moulds	3 per mix	Max load; compressive stress
Flexural behaviour	ASTM D790-17	180 × 60 × 20 mm, 120 mm span	3 per mix	Flexural strength; modulus
Reaction to fire	ISO 11925-2:2020	40 × 50 mm, 45°, 15 s	2	Time to ignition; to extinguish
Water absorption	ASTM D570-98	40 × 50 mm, shellac-coated	1	Mass gain, 24 h immersion
Bulk density	Gravimetric	50 mm cube	1	Mass / mould volume

REPORTING CONVENTION

Mechanical values are means of three specimens; standard deviations cannot be reported. Fire, absorption and density rest on single determinations and are indicative, not characteristic, values.

All testing: Advanced Construction Materials Research Laboratory, Hacettepe University.

Compressive strength scales with solids concentration

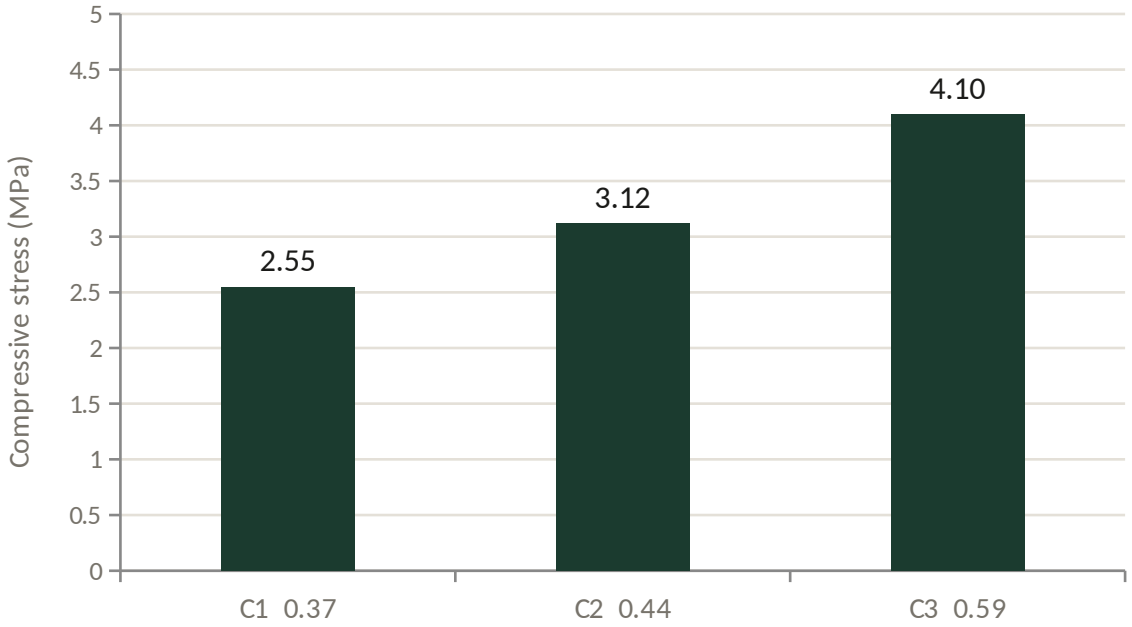
ID	Solids : water	F_{max} kN	σc MPa
C1	0.37	3.89	2.55
C2	0.44	4.30	3.12
C3	0.59	**6.80**	**4.10**

OBSERVED RELATIONSHIP

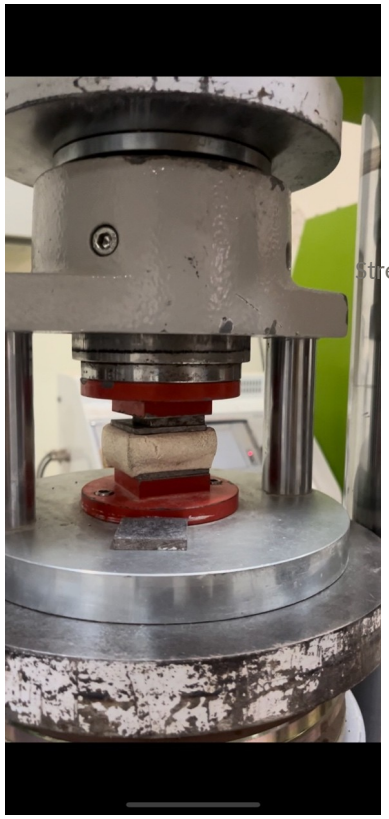
$$\sigma c = 6.93 \cdot (\text{solids : water}) + 0.04$$

$R^2 = 0.994$ over 0.37 – 0.59

With shell : binder fixed, strength is linear in solids concentration across the range tested. Three formulations cannot establish a law — but the reading is that packing density, not mineral content, sets strength.

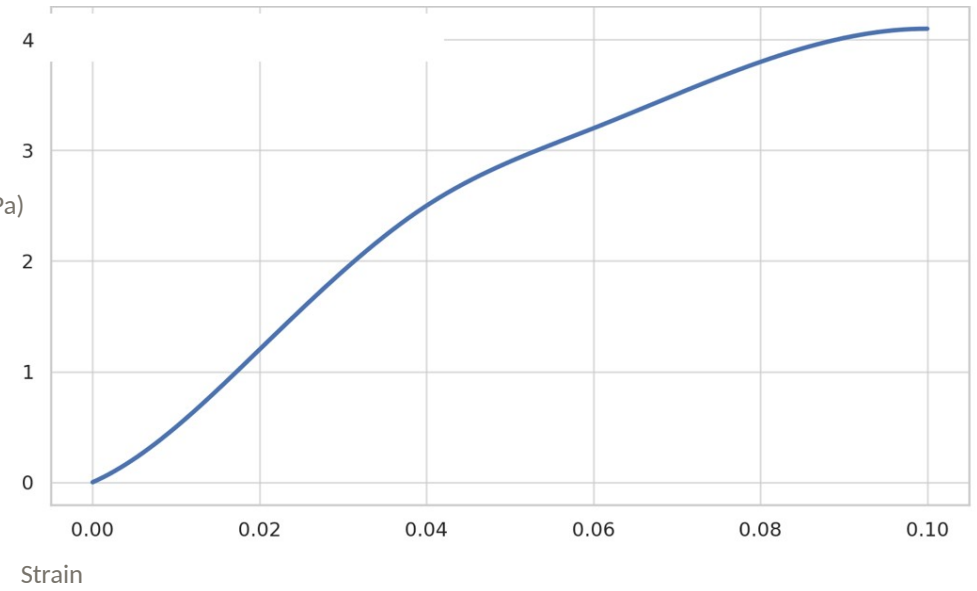


Failure mode: progressive crushing



C3 after loading: barrelled, shortened, intact

COMPRESSIVE STRESS – STRAIN, C3



1.38 g/cm³ · 2.96 MPa per g/cm³

Bulk density on a mould-volume basis; specific compressive strength.

READING THE CURVE

- Linear elastic to approximately 1 MPa
- Progressive stiffness loss as the agar network yields
- Peak at roughly 10% strain, no abrupt load drop
- Beyond peak: deformation with retained integrity

Strain derived from an assumed 10% maximum deformation based on observation, not instrumented.

Flexure: plasticiser content is decisive

ID	Glycerol % solids	F_{max} N	σF MPa	EF MPa	ε peak
F1	18.8	—	*not measurable*	—	—
F2	10.3	310	2.325	192	0.0121

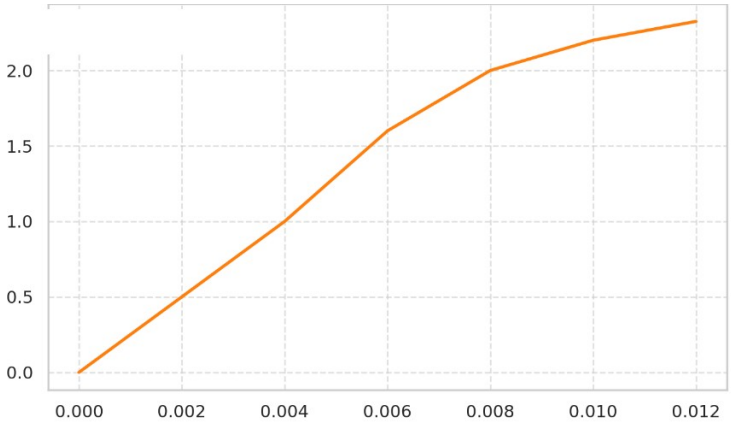
$\sigma_F = 3FL / 2bd^2$ with F = 310 N, L = 120 mm, b = 60 mm, d = 20 mm





Three-point rig; kerfed panel arched by hand

FLEXURAL STRESS – STRAIN, F2



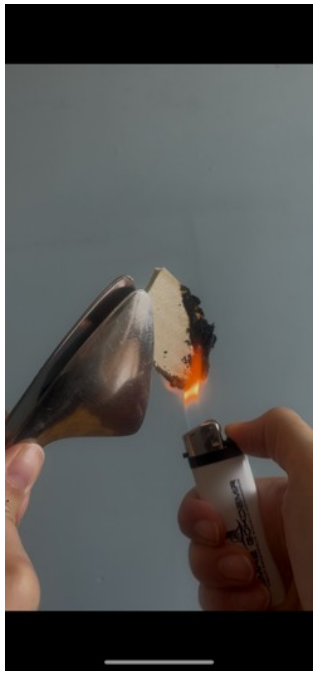
DISCLOSURE — SECTION GEOMETRY

The specimen carries transverse kerfs on its tension face. ASTM D790-17 assumes a solid rectangular section and the calculation uses the full 60 × 20 mm. Part of the measured compliance is geometric, so the modulus is a lower bound.

INTERPRETATION

At 18.8% glycerol the composite deformed without developing measurable resistance. Halving it produced a section that resists while still flexing. The transition lies between these values.

Fire, hygric response and density



IGNITABILITY — ISO 11925-2

~10 s
to ignition

4 s
to self-extinguish

- Charring local to flame contact
- No lateral flame spread
- No melting, no dripping

Probable mechanism: endothermic carbonate decomposition releasing CO_2. Not verified calorimetrically. Ignitability only — no Euroclass, cone calorimetry, smoke or toxicity data.

WATER ABSORPTION — ASTM D570

0.7%

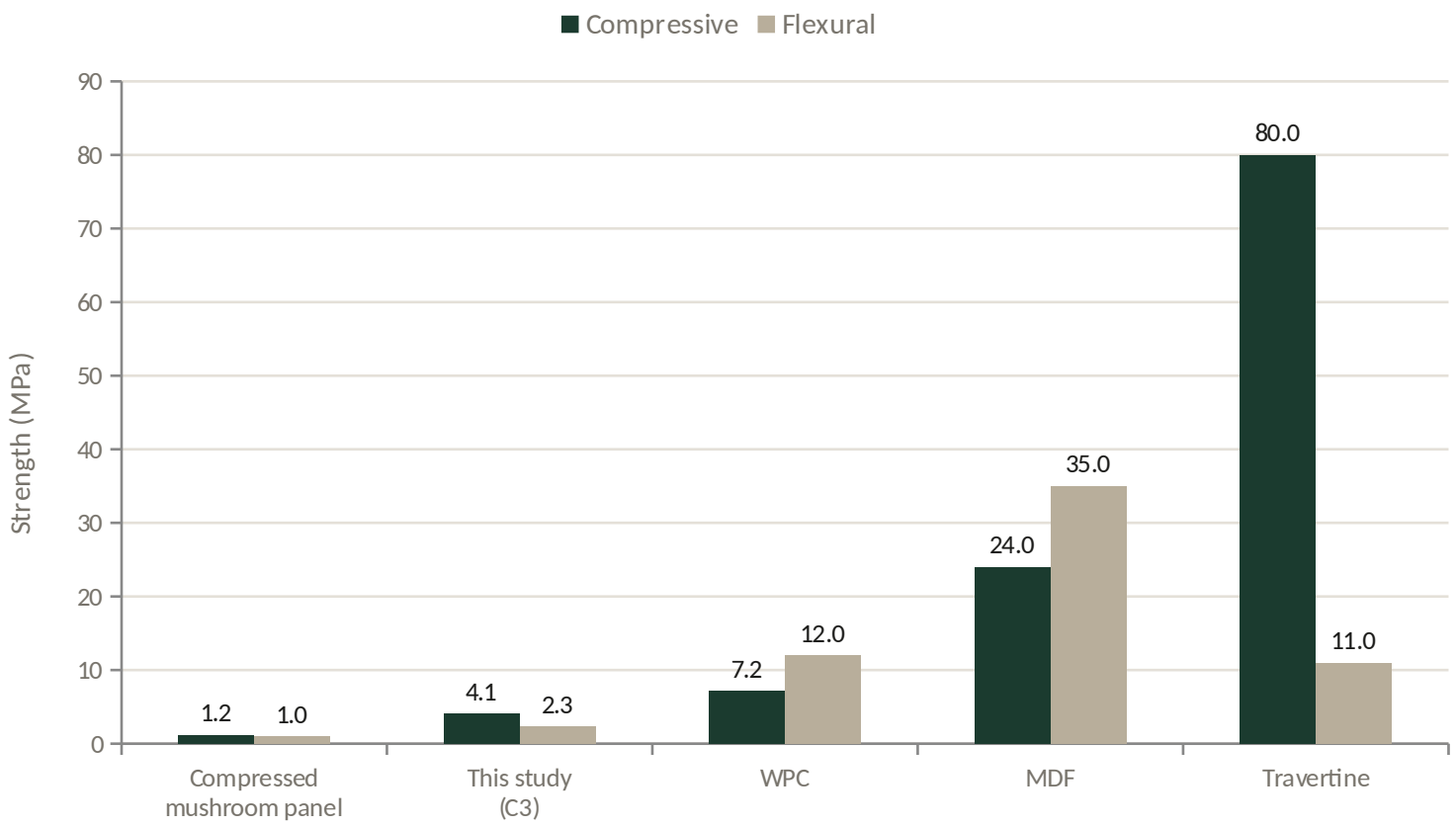
mass gain over 24 h immersion, shellac-coated. Local dissolution where the coating was thin: coating integrity is the limiting factor.

BULK DENSITY

Material	g/cm³	Reference
This study (C3)	**1.38**	—
Mycelium, cellulose	0.3 – 0.8	Jones et al. 2020
Fired clay brick	1.6 – 2.0	William & Owen 1995
Normal-weight concrete	2.4	William & Owen 1995

Quoted on a mould-volume basis and therefore a lower bound.

Benchmark I — conventional interior materials



Velasco & Gaona 2011 · Klyosov 2007 · Nemli & Aydın 2007 · Altun 2006

POSITION

One sixth the compressive strength of MDF and one twentieth that of travertine — a stone the material visually resembles. Three times compressed mushroom panel, the only other fully bio-based board here.

On strength alone it does not compete with engineered wood or stone, and this paper does not claim it does.

Benchmark II — bio-based composites

COMPRESSIVE STRENGTH

Material	σc MPa	Reference
This study (C3)	**4.1**	—
Mycelium composite	0.145 – 0.452	Jones et al. 2020
Hempcrete	0.3 – 0.6	Arnaud & Gourlay 2012
Eggshell-filled PLA	2 – 5	Kim & Pal 2010
Mineral-filled biocomposite	2.1 – 39.6	Shah et al. 2016
30% algae-filled	21.3	Kim & Pal 2010

ELASTIC MODULUS

Material	E MPa	Reference
This study (F2)	**192**	—
Mycelium composite	1.8 – 5	Islam et al. 2018
Soy-based bioplastic	124	Mansour et al. 2021
Corn starch + hemp	158	Dweib et al. 2004
Flax-fibre PLA	198	Faruk et al. 2012
LDPE	100 – 300	Zhao et al. 2022
Wood (pine)	11 000	Pope & Hudson 1989

An order of magnitude above the fungal and lime-binder family in compression, and inside the bio-polymer band in stiffness.

A water-cast, ambient-dried material reaches the stiffness of melt-processed bio-polymers.

Unfilled agar films in the packaging literature sit well below this: the mineral phase, not the binder, carries the stiffness.

A derived property: modulus of resilience

DEFINITION

$$U_r = \sigma_F^2 / 2E_F$$

Elastic strain energy absorbed per unit volume up to permanent deformation — the area under the curve to yield.

$$2.325^2 / (2 \times 192) = 14.1\ \mathrm{kJ/m^3}$$

from this study's own measured σ_F and E_F for F2

Stiffness sits in the denominator. The low modulus reported as a deficiency in Benchmark I is what raises this quantity.

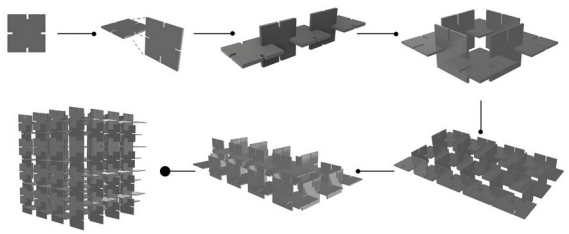
WHY IT FITS THIS CLASS

Both programmes gave the same qualitative result: substantial deformation with retained integrity. Strength records the load at which a material stops performing, not whether it then fails gracefully or catastrophically — which, for a demountable non-loadbearing component, governs service life.

PROPOSAL, AND ITS LIMIT

Report σ and E from the same specimen set so Ur can be derived. At present the literature reports them in separate studies on different formulations, so a comparative table cannot be assembled. This is a data point and a measurement convention, not a ranking.

Application pathway





DERIVED FROM THE MEASURED PROPERTIES

Low strength, high deformation capacity → self-weight and handling loads only; no structural duty

Self-extinguishing, no dripping → interior surfaces away from ignition sources

Coating-dependent water resistance → dry interiors; not wet rooms or facades

Repeatable small-format casting → a module, not a sheet — failure is replaced one unit at a time

SYSTEM CHARACTERISTICS

- Reversible assembly — cast units joined without adhesive, recoverable intact
- Distributed load path — units interlock and share load across neighbours
- Unit replaceability — a damaged unit is exchanged without dismantling
- Formulation-family variation — colour and texture from pigment and mould geometry

Potentials for sustainable interiors

Waste valorisation at source

A local food-waste stream becomes a surface material with no intermediate industry.

Low-energy production

No oven, no autoclave, no extrusion. One 55-minute cook, then ambient drying.

End-of-life by design

Every constituent biodegrades, including the shellac finish.

Reversible assembly

Adhesive-free modular assembly means reuse precedes recycling.

Displacing a mineral extraction

Substitutes quarried limestone in a surface application, not a downcycled use.

Low-barrier manufacture

A grinder, a sieve, a hotplate and silicone moulds. Production can be distributed.

No life-cycle assessment was carried out. These are qualitative potentials grounded in the protocol, not quantified environmental claims.

Limitations

Dispersion not reported

Means of three specimens; individual values not retained, so no standard deviations.

Formulation confound in flexure

Glycerol, solids concentration and shell : binder moved together between F1 and F2.

Section geometry in flexure

A kerfed specimen treated as a solid rectangle. The modulus is a lower bound.

Strain not instrumented

Compressive strain derived from an assumed 10% maximum deformation.

Shrinkage unquantified

Post-drying dimensions not logged; density is a lower bound.

Process not instrumented

Manual stirring; single laboratory, single operator, no inter-laboratory reproduction.

No durability data

No humidity or thermal cycling, UV exposure, microbial resistance or accelerated ageing.

No fire classification or LCA

Ignitability only; no Euroclass determination and no life-cycle assessment.

Each defines a specific next experiment rather than a general caveat.

Conclusions

01 **A reproducible protocol**

120 °C shell treatment, 150 µm fraction, 95 °C dissolution, 80 °C cook, 55 min, silicone casting, 11 days ambient.

02 **Solids concentration governs strength**

With shell : binder fixed at 2.00, compressive stress is linear in solids concentration across 0.37–0.59.

03 **Plasticiser sets the brittle–ductile transition**

Between 18.8% and 10.3% glycerol, from unmeasurable compliance to 2.33 MPa at 192 MPa.

04 **Bio-polymer stiffness, oven-free**

192 MPa places a water-cast, ambient-dried material inside the melt-processed bio-polymer band.

05 **Strength is the wrong sole criterion**

Deformation capacity governs service behaviour. σ and E should be reported together.

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Testing: Advanced Construction Materials Research Laboratory, Hacettepe University, with thanks to Dr Hüseyin İlcan. Derived from the author's doctoral dissertation, Hacettepe University, 2025.

Selected references

REVIEWS AND STATE OF THE ART

- Ngayakamo, B. & Onwualu, A.P. (2022). Valorisation of eggshell waste for sustainable construction materials. Heliyon.
- Owuamanam, S. & Cree, D. (2020). Progress of bio-calcium carbonate waste eggshell and seashell fillers in polymer composites. J. Composites Science, 4(2), 70.
- Biogenic calcium carbonate from waste shells for advanced material applications (2022). Frontiers in Materials, 9, 1024977.

AGAR AND BIO-POLYMER SYSTEMS

- Hernández, V. et al. (2022). Agar biopolymer films for biodegradable packaging: a reference dataset. Materials, 15(11), 3954.
- Materiom (2026). Open biomaterial recipe commons and property database. materiom.org/data.
- Mansour et al. (2021); Dweib et al. (2004); Faruk et al. (2012); Zhao et al. (2022) — bio-polymer elastic moduli.

COMPARATIVE MATERIAL DATA

- Jones, M. et al. (2020); Islam et al. (2018) — mycelium composites. Arnaud & Gourlay (2012) — hempcrete.
- Kim & Pal (2010) — eggshell-filled PLA. Shah et al. (2016) — mineral-filled biocomposites.
- Nemli & Aydın (2007) — MDF. Altun (2006) — travertine. Klyosov (2007) — WPC. Velasco & Gaona (2011) — mushroom panel.

Standards: ASTM D790-17 · ISO 11925-2:2020 · ASTM D570-98(2018).